

Max Weight Perfect Matching

$$\max: \sum w_e x_e$$

$$\text{st: } x_e \geq 0$$

$$x(\delta(v)) = 1$$

$$x(\delta(u)) \geq 1, \quad |u| \geq 3, \text{ odd}$$

(assume $\exists$ a perf. matching).

$$\min: \sum y_u$$

$$\text{st: } \sum_{u: e \in \delta(u)} y_u \leq w_e$$

$$e \in \delta(u)$$

$$y_u \geq 0, \quad \forall u \in V, \quad \underline{|u| \geq 3} \quad \text{odd.}$$

Primal has exp^y many constraints

Dual " " " " variables

Limiting Variables in Dual

Idea: Build a dual solution
with only a few non-zero entries

Namely: maintain a laminar
family $\left[\begin{array}{c} \text{two nested ellipses with a dot in the inner one} \\ \text{two separate ellipses with dots in each} \end{array} \right]$

Ω and only consider dual
variables indexed by some $u \in \Omega$.

Check: $|\Omega| = O(|V|)$

Dual Variables and CS

Maintain laminar family Ω

Consider only y_u , $u \in \Omega$.

If $u \in \Omega$: y_u is allowed to
be +ve (but can be zero).

Alg will update y in Ω .

CS : $\sum_{\substack{u: \\ e \in u}} y_u \leq w_e$ For y dual-feasible

$E_y = \{e : w_e - \sum_{u \in \Omega} y_u = 0\}$

RP can only use these edges.

Primal-Dual Framework — 1/2

y dual feasible.

If we can find x that is ^{or a matching}

primal feasible & satisfies CS $\Rightarrow$ done!

If not: use failure to improve y .

For max wt matching: Restricted Primal (RP) searches for any perf. matching using only edges in E_y .

If we find such a perf matching, it must be max wt:

$$w(M) = \sum_{e \in M} w_e \geq \sum_{e \in M} \sum_{u \in e} y_u = \sum_u y_u \cdot |M \cap \delta(u)| \geq \sum_u y_u$$

$\uparrow$ CS " $=$ "

$\uparrow$ " $=$ " b/c M is perfect

Primal-Dual Framework — 2/2

Alg will :

search for perf. matching
using Blossom algorithm.

Keep track of

- blossoms
- M -alt trees

Update Ω & improve the dual.